



**IT 112 Web Systems and Technologies
BSIT-2B 2nd Semester**

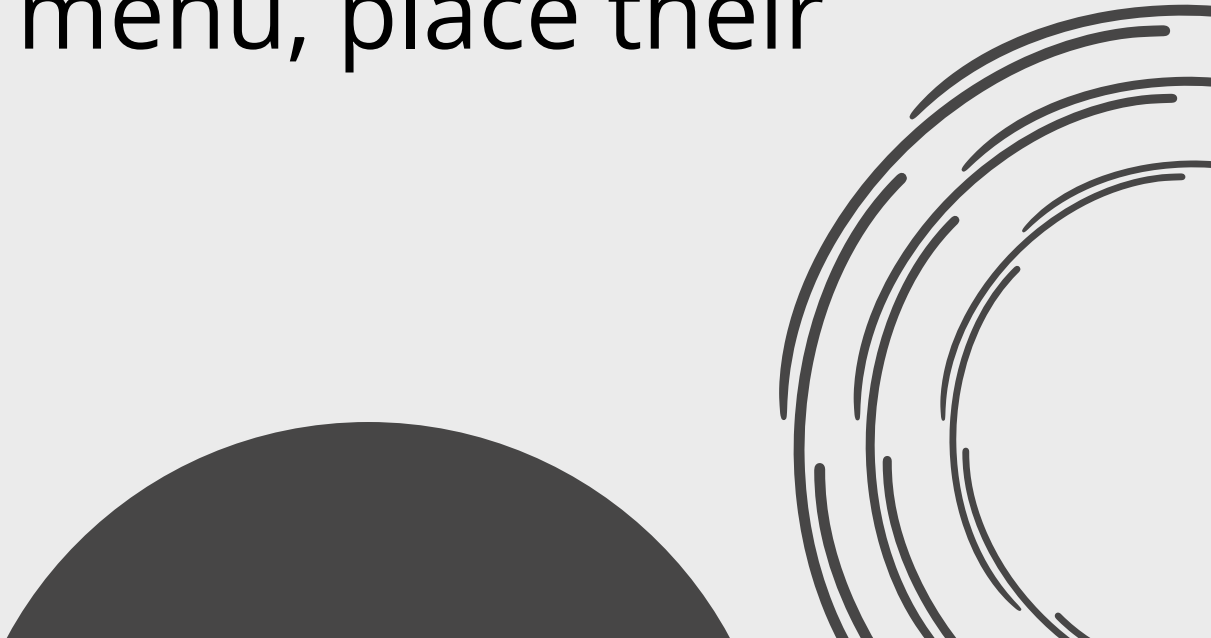
Crumbelle Ordering System





PROJECT INTRODUCTION

The Crumbelle Ordering System is our proposed way of making restaurant ordering easier, faster, and more organized. We noticed that in a busy restaurant setup, taking orders manually can sometimes cause delays and mistakes, which can affect both customers and staff. With this system, we want to make things more convenient by letting customers easily browse the menu, place their orders, and track them without hassle.





System Purpose

Our system aims to simplify the ordering process by providing a digital platform where users can conveniently order products and track their purchases.

Target Users

Customers

Admin (Store Manager)





TECHNOLOGY STACK

Frontend Technologies

The Frontend is responsible for creating the visual interface and user interface of the system.

- HTML - Used for structuring the content and layout of the web pages.
 - CSS - Used for designing the interface, styling components and creating a design.
 - JavaScript - Used to add interactivity, dynamic content, and frontend functionalities.
- 



Backend Technologies

Backend handles the system logic, API processing, and communication between the frontend and database

- Node.js - serve as the runtime environment for executing server-side JavaScript.
 - Express.js - A lightweight used for building API routes and handling server request and responses.
 - JavaScript - Used for backend logic, controllers, authentication, and server-side operations.
- 



Data-Base Technologies

The database manage the storage and retrieval of application data.

- MongoDB - A NoSQL database used to store customer information, product details, and order records.
 - JavaScript models and routes - Used for creating schemas, managing database operation, and handling application routing.
 - .env File - Stores sensitive configuration data such as database connection URLs, API keys, and secret credentials securely.
- 



SYSTEM FEATURES OVERVIEW

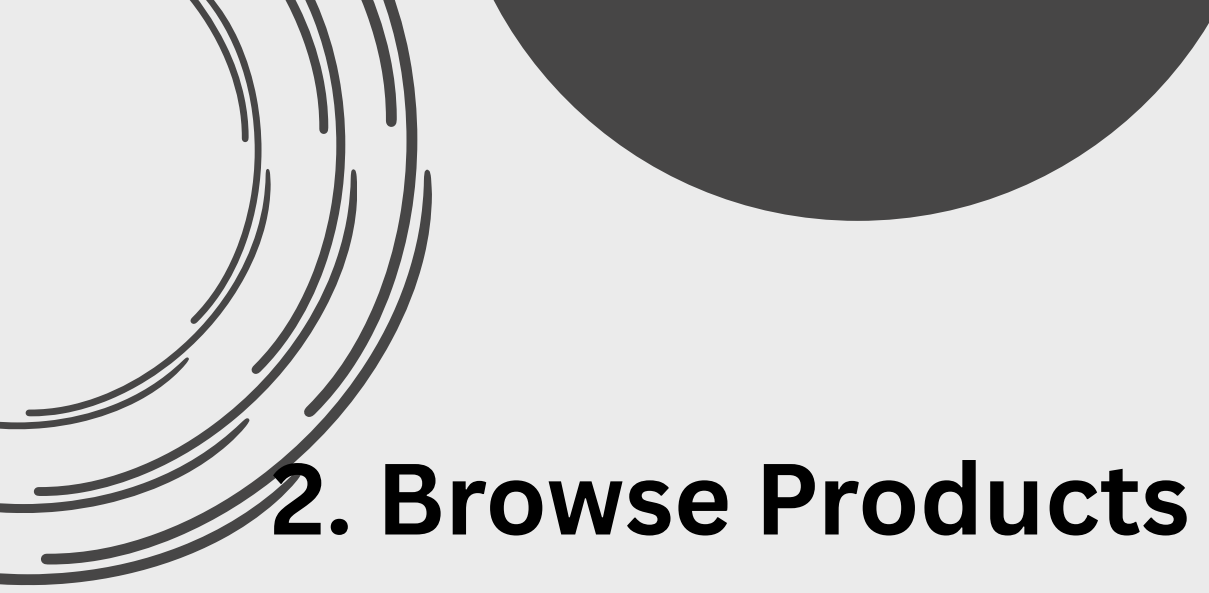
1. User Registration and Login (JWT Authentication)

The system allows users to create accounts and securely log in using JWT (JSON Web Token) authentication. This feature ensures that user information and transactions are protected while providing secure access to customer and admin accounts.

Functions:

- User sign-up and login
- Secure authentication using JWT
- Account session management
- Role-based access for customers and admins





2. Browse Products and View Details

Customers can easily browse available products and view detailed information before placing an order. This helps users make informed purchasing decisions.

Functions:

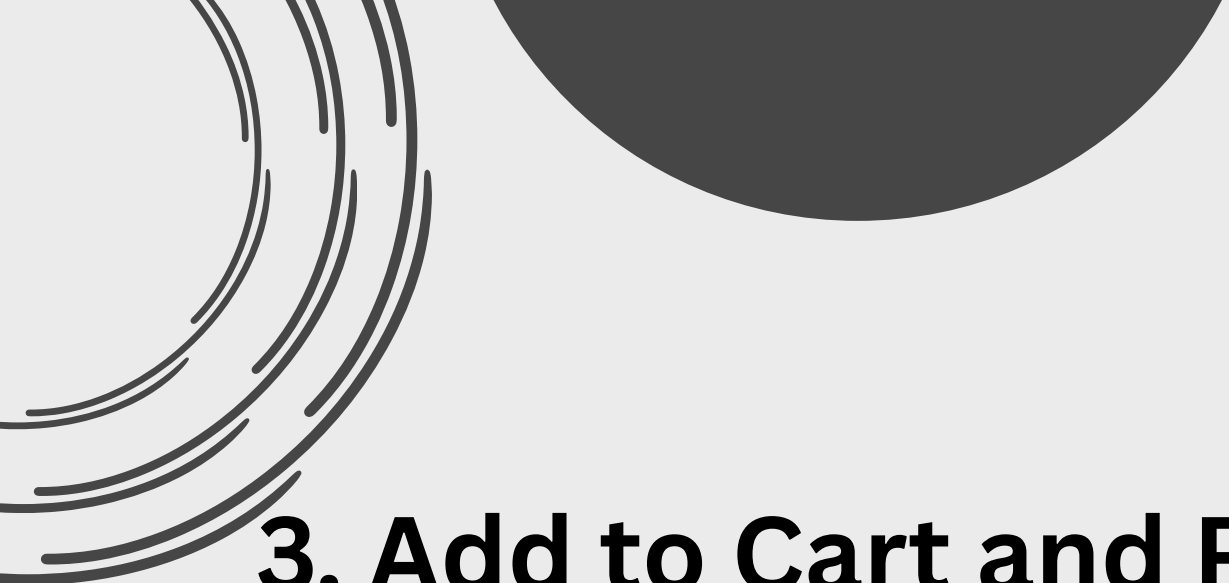
Display product listings

View product images and descriptions

Check prices and availability

Search and browse products conveniently





3. Add to Cart and Place Orders

The system provides a shopping cart feature where customers can add products, review their selections, and place orders quickly.

Functions:

- Add and remove items from cart
 - Update item quantities
 - Calculate total order cost
 - Confirm and place orders online
- 



4. Order Management (Admin Side)

Administrators can efficiently manage customer orders through the admin dashboard. This feature helps businesses organize and monitor transactions.

Functions:

View customer orders

Update order status

Track pending and completed orders

Manage order processing efficiently





5. Responsive User Interface

QuickServe is designed with a responsive interface that works smoothly on desktops, tablets, and mobile devices. The layout adapts to different screen sizes to provide a better user experience.

Functions:

Mobile-friendly design

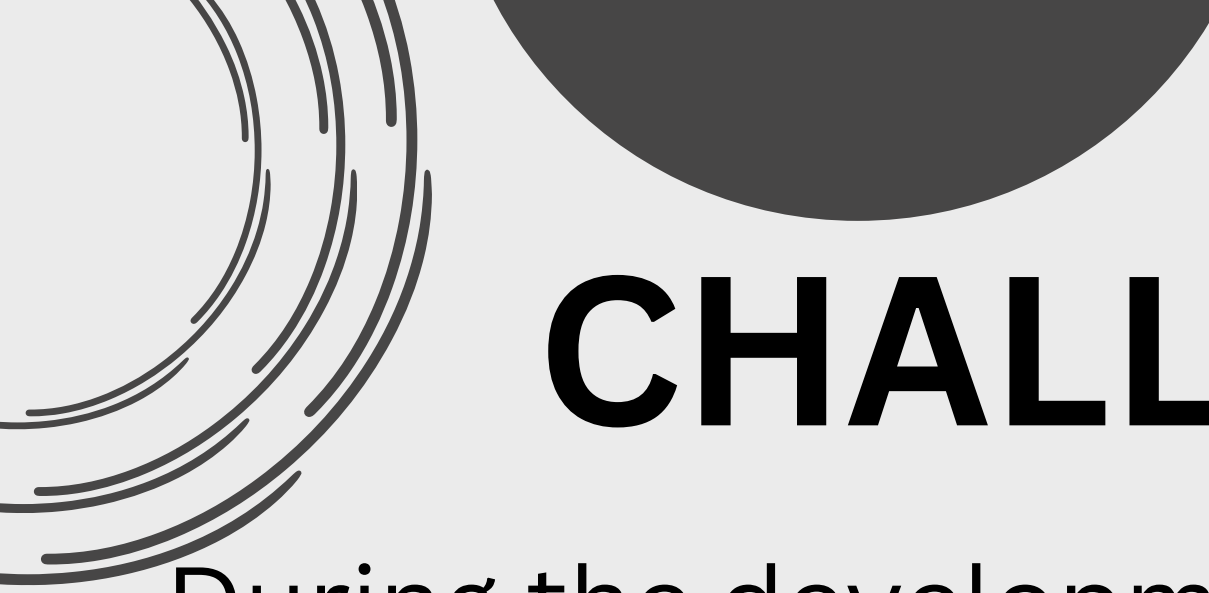
Easy navigation Interactive dashboard

Fast and user-friendly interface





LIVE DEMONSTRATION



CHALLENGES ENCOUNTERED

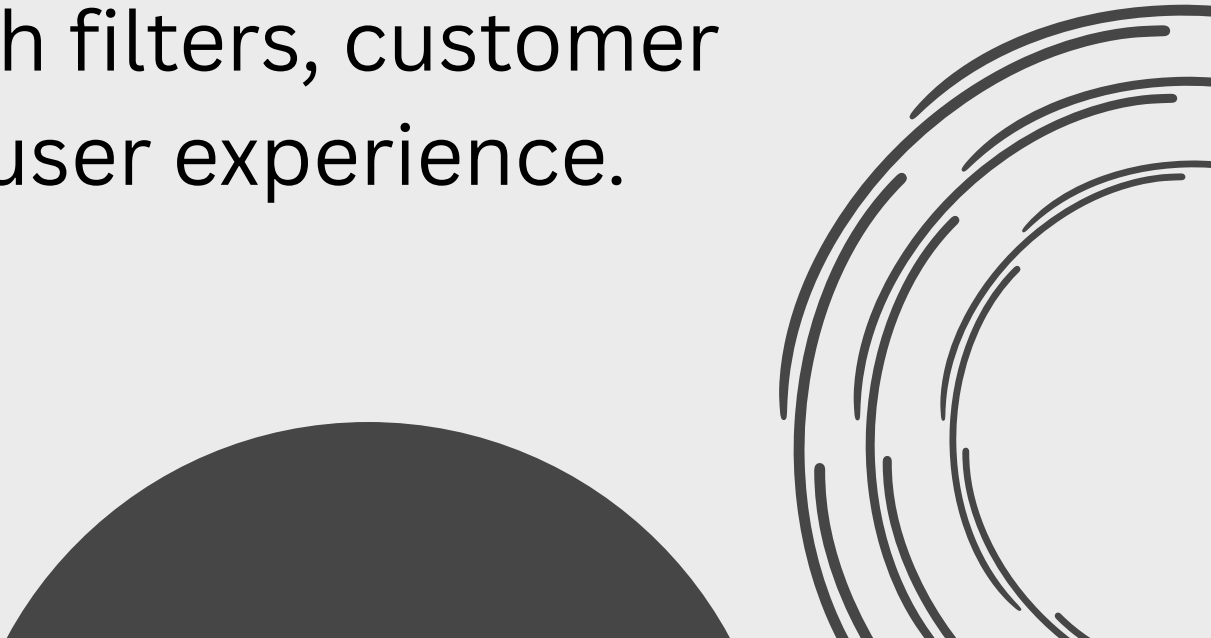
During the development of the QuickServe Ordering System, the team encountered several technical challenges such as sorting and ordering system issues, images not displaying properly, and coding errors. Problems also occurred when connecting MongoDB due to incorrect .env configurations and database connection settings. And the API testing in Thunder Client, incorrect terminal directories, and MongoDB network whitelisting caused further difficulties that required continuous debugging and testing to resolve.





FUTURE IMPROVEMENTS

The QuickServe Ordering System can be enhanced by adding more advanced features and improving overall performance. The system can include real-time order notifications through email or SMS to keep customers updated on their orders. An improved admin dashboard with sales reports, analytics, and inventory management can help monitor business operations better. The user interface can also be upgraded with a more modern design, better animations, and mobile responsiveness. Stronger security measures can be added to protect user data and improve system safety. The system can be optimized to handle more users and products for better scalability. Additional features like search filters, customer reviews, and order history tracking can further improve the user experience.





Q & A





MEMBERS

Name	Role
Samuel L. Gonzales	(Backend Developer, Project Manager)
Roel B. Moendem	(Frontend Developer)
Nicka D. Tambobong	(Tester & Debugger, Documentation Specialist)
Maicel Joy Sadia	(GitHub Manager)
Aedan Jaden Pielago	(Database Manager)

